

1	Course Name:
	Cryptography and Steganography
2	Course Code:
	IKB21303
3	Year and Semester Offered:
	Year 1 Semester 3
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	IKB10403 IT Security Essentials
6	Course Synopsis: This course covers the three domain of digital information hiding that consists of cryptography, steganography and digital watermarking. Current trends of the digital information hiding approaches and practices are introduced to protect sensitive information in a communication over the Internet especially during the presence of adversarial behaviour. The overview for cryptography includes symmetric, asymmetric, cryptographic hash functions, key management and digital signature. Then, the outline for steganography includes hiding information within digital text document, digital images, digital audio and digital video. Lastly, digital watermarking encompasses current approaches of embedding a digital watermark into a digital media such as image, audio, or video. At end of the course, students are expected to apply the existing approaches in the real-world implementation.
7	Course Learning Outcome: Upon completion of the course, students should be able to: 1. Describe the essentials of cryptography, digital steganography and digital watermarking to protect confidential information. 2. Identify how to implement the components of cryptography, digital steganography and digital watermarking in an application. 3. Build an application based on the current approaches of digital information hiding.
8	Content Outline:
	Topic
	1. Introduction to Digital Information Hiding
	2. Cryptography (Part 1)
	3. Cryptography (Part 2)
	4. Cryptography (Part 3)
	5. Cryptography (Part 4)
	6. Image Steganography
	7. Audio and Video Steganography
	8. Text Steganography
	9. Digital Watermarking for Image
	10. Digital Watermarking for Audio

9	References
	David Megías, Wojciech Mazurczyk, Minoru Kuribayashi (2022). Data Hiding and Its Applications: Digital Watermarking and Steganography, ISBN-10 : 3036529365, Mdpi AG.